# Yang Du | CS

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Final year undergraduate at U-M pursuing a major in Computer Science and minor in Mathematics, applying for Ph.D. in Computer Science. Research interests: theoretical computer science, cryptography, blockchain.

## Education

**University of Michigan (Ann Arbor)** — Michigan, United States
B.S.E. in Computer Science — Major GPA: 4.0/4.0 — *Sep, 2019 - May, 2021*

**Shanghai Jiao Tong University** — Shanghai, China
B.S.E. in Electrical and Computer Engineering — Major GPA: 3.88/4.0 — *Sep, 2017 - Aug, 2019*

## Research Projects

**Oblivious Data Structure**
*Guide: Prof. Elaine Shi, CS Dept, Carnegie Mellon University* — *Apr, 2020 - Present*

- Implement an oblivious AVL tree data structure with Path ORAM and deterministic eviction.
- Find and compare efficiency of oblivious range query data structure.

**Efficient Algorithm to find number of prime factors of Integer with Oracle**
*Guide: Dr. Ilya Volkovich, CSE Dept, University of Michigan* — *April, 2020 - Sep, 2020*

- Devise the first efficient deterministic algorithm for approximating $\omega(N)$ the number of prime factors of an integer given in addition oracle access to Euler's Totient function.
- Show that the algorithm can be extended to handle a more general class of additive functions that "depend solely on the exponents in the prime factorization of an integer"
- Gives the first algorithm that approximates $\omega(N)$ without necessarily factoring $N$.

**Secure Cloud Manufacturing**
*Guide: Prof. Z. Morely Mao, EECS Dept, University of Michigan* — *Sep, 2020 - Present*

- Cybersecurity implementation, cloud system design, database creation and management, data analytics and verification

**Free Speech Technology**
*Guide: Prof. Dick Robert, EECS Dept, University of Michigan* — *Dec, 2019 - May, 2020*

- Designed to support free speech by preventing blocking, censorship, and surveillance.
- Support communication during events that disable the internet and cellular communication infrastructures.

## Publication

**Yang Du**, Ilya Volkovich, ***Approximating the Number of Prime Factors Given an Oracle to Euler's Totient Function***, ITCS 2021, submitted.

## Key Academic Projects

**Cryptocurrency and Blockchain**
*Guide: Prof. Mahdi Cheraghchi, CSE Dept, University of Michigan* — *Mar, 2020 - Apr, 2020*

- Study Bitcoin (public key encryption) and Z-cash (zero-knowledge proof).
- Blockchain application to e-voting and supply chain tracking.

**Deep Learning Techniques for Classification and Feature Learning**
*Guide: Dr. Kutty Sindhu Krishnan, CSE Dept, University of Michigan* — *Feb, 2020 - Apr, 2020*

- Use Pytorch to classify pictures different types of food.
- Compare different methods: NN, CNN, Autoencoder.

## Teaching Experience and Internship

**Linear Algebra, Proof Tutor**
*Instructor: Dr. Scott Schneider, Maths Dept, University of Michigan* *Jan, 2020 - May, 2020*
- Hold office hours to discuss homework questions with students.

**Foundation of Computer Science, Grader**
*Instructor: Dr. Amir Kamil, CSE Dept, University of Michigan* *Sep, 2019 - May, 2020*
- Grade homework and exam paper.

**Honors Mathematics II, Teaching Assistant**
*Instructor: Dr. Horst Hohberger, Shanghai Jiao Tong University* *Sep, 2019 - May, 2020*
- Lead weekly discussion session.
- Grade homework and exam paper.

**Software Engineer of Developing a Teaching Management System**
*Shanghai Suxun Education Ltd.* *Dec, 2018 - Feb, 2019*
- Front end(Android) development.
- Design and format API and design data structure to minimize data transferred.

## Scholastic Achievements

- **Roger King Scholarship** awarded by University of Michigan *Aug, 2020*
- **Dean's Honors List** awarded by University of Michigan *Dec, 2019*
- **Fuda Scholarship** awarded by Shanghai Jiao Tong University *Nov, 2018*
- **Undergraduate Merit Scholarship** awarded by Shanghai Jiao Tong University *Nov, 2018 & 2019*

## Technical Strengths

- **Programming Languages:** C/C++, Python, Java
- **Development:** Android
- **Others:** LaTeX, Matlab, Mathematica, Origin